Binomial Distribution

Example 1: If a coin is tossed 5 times, find the probability of:

(a) Exactly 2 heads

(b) At least 4 heads.

Solution:

Number of trials: n=5

Probability of head: p= 1/2 and hence the probability of tail, q =1/2

For exactly two heads:

x=2

$P(x=2) = {}^{5}C2\ p^{2}\ q^{5-2} = 5!\ /\ 2!\ 3! \times (½)^{2} \times (½)^{3}$

P(x=2) = 5/16

(b) For at least four heads,

$x \geq 4$, $P(x \geq 4) = P(x = 4) + P(x=5)$

Hence,

$P(x = 4) = {}^{5}C4\ p^{4}\ q^{5-4} = 5!/4!\ 1! \times (½)^{4} \times (½)^{1} = 5/32$

$P(x = 5) = {}^{5}C5\ p^{5}\ q^{5-5} = (½)^{5} = 1/32$

Therefore,

$P(x \geq 4) = 5/32 + 1/32 = 6/32 = 3/16$

Example 2: For the same question given above, find the probability of:

a) **Getting at most 2 heads**

Solution: P (at most 2 heads) = P(X ≤ 2) = P (X = 0) + P (X = 1) + + P (X = 2)

P(X = 0) = $(½)^5$ = 1/32

P(X=1) = $^5C_1$ $(½)^5$ = 5/32

P(x=2) = $^5C2$ p^2 q^{5-2} = 5! / 2! 3! × $(½)^2$× $(½)^3$ = 5/16

Therefore,

P(X ≤ 2) = 1/32 + 5/32 + 5/16 = ½

Example

If X is binomially distributed with 6 trials and a probability of success equal to $\frac{1}{4}$ at each attempt, what is the probability of:

(a) exactly 4 successes (b) at least one success?

Solution:

(a)
$$P(X=4) = \binom{6}{4}\left(\frac{1}{4}\right)^4\left(\frac{3}{4}\right)^2$$
$$= 15 \times \frac{1}{256} \times \frac{9}{16}$$
$$= \frac{135}{4096} = 0.033 \quad \text{(to 3 d.p.)}.$$

(b)
$$P(X \geq 1) = 1 - P(X=0)$$
$$= 1 - \left(\frac{3}{4}\right)^6$$
$$= 1 - \frac{729}{4096}$$
$$= \frac{3367}{4096} = 0.822 \quad \text{(to 3 d.p.)}.$$